8.6 Addition Theorem of Probability

(i) If A and B are any two events then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

(ii) If A, B and C are any three events then

$$P\left(A \cup B \cup C\right) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

Proof

(i) Let A and B be any two events of a random experiment with sample space S.

From the Venn diagram, we have the events only A, $A \cap B$ and only B are mutually exclusive and their union is $A \cup B$

$$\begin{aligned} \text{Therefore, } P(A \cup B) &= P[\text{(only A)} \cup (A \cap B) \cup \text{(only } B)] \\ &= P(\text{only } A) + P(A \cap B) + P(\text{only } B) \\ &= [P(A) - P(A \cap B)] + P(A \cap B) + [P(B) - P(A \cap B)] \\ P(A \cup B) &= P(A) + P(B) - P(A \cap B) \end{aligned}$$

(ii) Let A, B, C are any three events of a random experiment with sample space S.

$$\begin{aligned} \text{Let} \qquad D &= B \cup C \\ P(A \cup B \cup C) &= P(A \cup D) \\ &= P(A) + P(D) - P(A \cap D) \\ &= P(A) + P(B \cup C) - P[A \cap (B \cup C)] \\ &= P(A) + P(\text{B}) + P(C) - P(B \cap C) - P[(A \cap B) \cup (A \cap C)] \\ &= P(A) + P(B) + P(C) - P(B \cap C) - P(A \cap B) - P(A \cap C) + P[(A \cap B) \cap (A \cap C)] \\ P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) \\ &\quad - P(C \cap A) + P(A \cap B \cap C) \end{aligned}$$

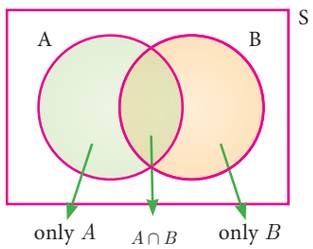


Fig. 8.10

Activity 5

The addition theorem of probability can be written easily using the following way.

$$\begin{aligned} P(A \cup B) &= S_1 - S_2 \\ P(A \cup B \cup C) &= S_1 - S_2 + S_3 \end{aligned}$$

Where $S_1 \rightarrow$ Sum of probability of events taken one at a time.

$S_2 \rightarrow$ Sum of probability of events taken two at a time.

$S_3 \rightarrow$ Sum of probability of events taken three at a time.

$$P(A \cup B) = \underbrace{P(A) + P(B)}_{S_1} \quad \underbrace{-P(A \cap B)}_{S_2}$$

$$P(A \cup B \cup C) = \underbrace{P(A) + P(B) + P(C)}_{S_1} \quad \underbrace{-(P(A \cap B) + P(B \cap C) + P(A \cap C))}_{S_2} \quad \underbrace{+P(A \cap B \cap C)}_{S_3}$$

Find the probability of $P(A \cup B \cup C \cup D)$ using the above way. Can you find a pattern for the number of terms in the formula?

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